DESIGN OF A LOW-COST WIRELESS ECG SYSTEM FOR CONTINUOUS CARDIAC MONITORING USING BIOMEDICAL TRANSDUCERS

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

ECA 0710 – TRANDUCERS AND SENSORS FOR INDUSTRIAL AUTOMATION

to the award of the degree of

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

SUBMITTED BY

HARSHEKA PRADEEP(192412531)

VEESAM HARINI(192412566)

Under the Supervision of

Dr. SENTHILKUMAR K P



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

July 2025

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

DECLARATION

We, **HARSHEKA PRADEEP AND VEESAM HARINI** of the **ELECTRONICS AND COMMUNICATION ENGINEERING,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **'DESIGN OF A LOW-COST WIRELESS ECG SYSTEM FOR CONTINUOUS CARDIAC MONITORING USING BIOMEDICAL TRANSDUCERS'** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: THANDALAM, CHENNAI

Date:

Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Engineer to Excel

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **'DESIGN OF A LOW-COST WIRELESS ECG SYSTEM FOR CONTINUOUS CARDIAC MONITORING USING BIOMEDICAL TRANSDUCERS'** has been carried out by **HARSHEKA PRADEEP AND VEESAM HARINI** under the supervision of Dr SENTHILKUMAR K P and is submitted in partial fulfilment of the requirements for the current semester of the B.E in **ELECTRONICS AND COMMUNICATION ENGINEERING,** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Name of the Program Director

Program Director

Department Name

ELECTRONICS AND COMMUNICATION ENGINEERING

Saveetha School of Engineering

SIMATS

SIGNATURE

Name of the Guide

Dr SENTHILKUMAR K P

Designation

Department Name

ELECTRONICS AND COMMUNICATION ENGINEERING

Saveetha School of Engineering

SIMATS

Submitted for the Project work Viva-Voce held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

This capstone project presents the design and development of a low-cost, wireless Electrocardiogram (ECG) monitoring system aimed at providing continuous cardiac health surveillance using biomedical transducers. Cardiovascular diseases remain the leading cause of mortality worldwide, and early detection through continuous ECG monitoring can significantly improve outcomes. However, traditional ECG systems are either bulky or expensive, limiting their accessibility in resource-constrained environments. The proposed system integrates biomedical transducers to accurately capture the electrical activity of the heart and convert it into analog signals. These signals are then filtered, amplified, and digitized for further processing. A microcontroller processes the ECG data and transmits it wirelessly via Bluetooth or Wi-Fi to a smartphone or cloud server for real-time monitoring and analysis. The system emphasizes portability, affordability, and energy efficiency, making it suitable for home-based patient care and remote health monitoring. Key features include low power consumption, user-friendly interface, and reliable data transmission. The prototype was tested on multiple parameters such as signal clarity, transmission range, and battery performance, and results showed consistent and clinically interpretable ECG waveforms. The design also supports alert generation for abnormal heart activity, making it a useful tool for preventive healthcare.

ACKNOWLEDGMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, Program Director for his continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide, Dr SENTHILKUMAR K P for his creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

HARSHEKA PRADEEP(192412531)

VEESAM HARINI(192412566)

TABLE OF CONTENT

Chapter No	Content	Page No
1.	**INTRODUCTION**	**7**
	1.1 Background Information	7
	1.2 Project Objectives	7
	1.3 Significance of the Project	8
	1.4 Scope of the Project	8
	1.5 Methodology Overview	8
2.	**PROBLEM AND IDENTIFICATION ANALYSIS**	**9**
	2.1 Description of the Problem	9
	2.2 Evidence of the Problem	9
	2.3 Stakeholders Affected	9
	2.4 Supporting Data/Research	10
3.	**SOLUTION DESIGN AND IMPLEMENTATION**	**11**
	3.1 Development and Design Process	11
	3.2 Tools and Technologies Used	11
	3.3 Solution Overview	12
	3.4 Engineering Standards Applied	12
	3.5 Solution Justification	12
4.	**RESULTS AND RECOMMENDATIONS**	**13**
	4.1 Evaluation of Results	13

4.2 Challenges Encountered 13

4.3 Possible Improvements 14

4.4 Recommendations for Future Work 14

5. REFLECTION LEARING AND PERSONAL DEVELOPMENT 15

5.1 Key Learning Outcomes 15

5.2 Challenges Encountered and Overcome 15

5.3 Application of Engineering Standards 15

5.4 Insights into the Industry 15

5.5 Conclusion on Personal Development 16

6. CONCLUSION 17

6.1 Summary of Key Findings 17

6.2 Significance and Impact of the Project 17

7. REFERENCES 18

8. APPENDICES 19

TABLE OF FIG :

S.NO	FIGURE NAME	PAGE.NO
1.	FIG : 2.4.1	11
2.	FIG : 5.1	16

CHAPTER 1

INTRODUCTION

1.1 Background

Cardiovascular diseases are a leading global cause of death. Early detection and continuous monitoring are vital for effective treatment. Traditional ECG systems are costly and stationary, making them unsuitable for long-term use in remote or low-resource areas. The advancement of wearable tech and wireless communication enables the development of a low-cost, portable ECG system for real-time cardiac monitoring.

1.2 Objectives

- Design a low-cost, portable ECG system using biomedical transducers.
- Enable real-time cardiac signal acquisition and wireless data transmission.
- Visualize ECG signals on mobile or cloud platforms.
- Incorporate alerts for abnormal heart activity.
- Ensure usability in remote or resource-limited settings.

1.3 Significance

This project provides an affordable, accessible cardiac monitoring solution, particularly for underserved and rural populations. It supports home care, elderly monitoring, and telemedicine, helping reduce hospital visits and improve early diagnosis of heart conditions.

1.4 Scope

- Develop a prototype with biomedical transducers.
- Implement signal conditioning (filtering & amplification).
- Use a microcontroller for processing and wireless transmission.
- Create a basic mobile/PC interface to display ECG waveforms.

Exclusions

- No AI/ML diagnosis

- Not for clinical use or mass production
- No focus on battery life or waterproofing

Methodology

1. **Select Components** – Choose low-cost sensors, MCU, and wireless modules
2. **Build Circuit** – Design and test ECG signal conditioning
3. **Program MCU** – Capture and process ECG signals
4. **Enable Wireless** – Transmit data via Bluetooth/Wi-Fi
5. **Display Data** – Show ECG on phone/PC interface
6. **Test System** – Check signal quality and device reliability

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Problem Overview

Traditional ECG machines are costly, bulky, and clinic-bound—limiting access for patients in rural or low-income areas. Chronic heart patients lack affordable, portable, and real-time monitoring, leading to missed early warnings and delayed care.

2.2 Supporting Evidence

- WHO: 17.9 million CVD deaths annually; 75% in low/mid-income countries.
- India: CVDs cause ~28% of total deaths.
- Rural case studies show delayed diagnosis due to lack of portable ECGs—resulting in higher costs and preventable deaths.

2.3 Stakeholders

- Patients – Need affordable, home-based cardiac monitoring
- Healthcare Workers – Require portable tools for field diagnosis
- Clinics/Hospitals – Seek low-cost, efficient monitoring systems
- Public Health Agencies/NGOs – Focused on rural and remote healthcare delivery

2.4 Supporting Data/Research

Several studies and reports support the need for affordable ECG monitoring systems:

- A study published in The Lancet (2020) highlights the rising burden of CVDs in South Asia and the need for scalable diagnostic tools.
- Research from the Indian Heart Journal (2019) notes that portable ECG devices can reduce hospital admissions by 20–30% if integrated with early warning systems.
- A technical report by the National Health Mission (NHM) states that over 60% of Primary Health Centers in India lack ECG facilities due to cost and infrastructure constraints.

- WHO's Global Health Observatory reports suggest that integrating telemedicine and mobile health solutions can reduce diagnostic delays by up to 40% in low-income settings.

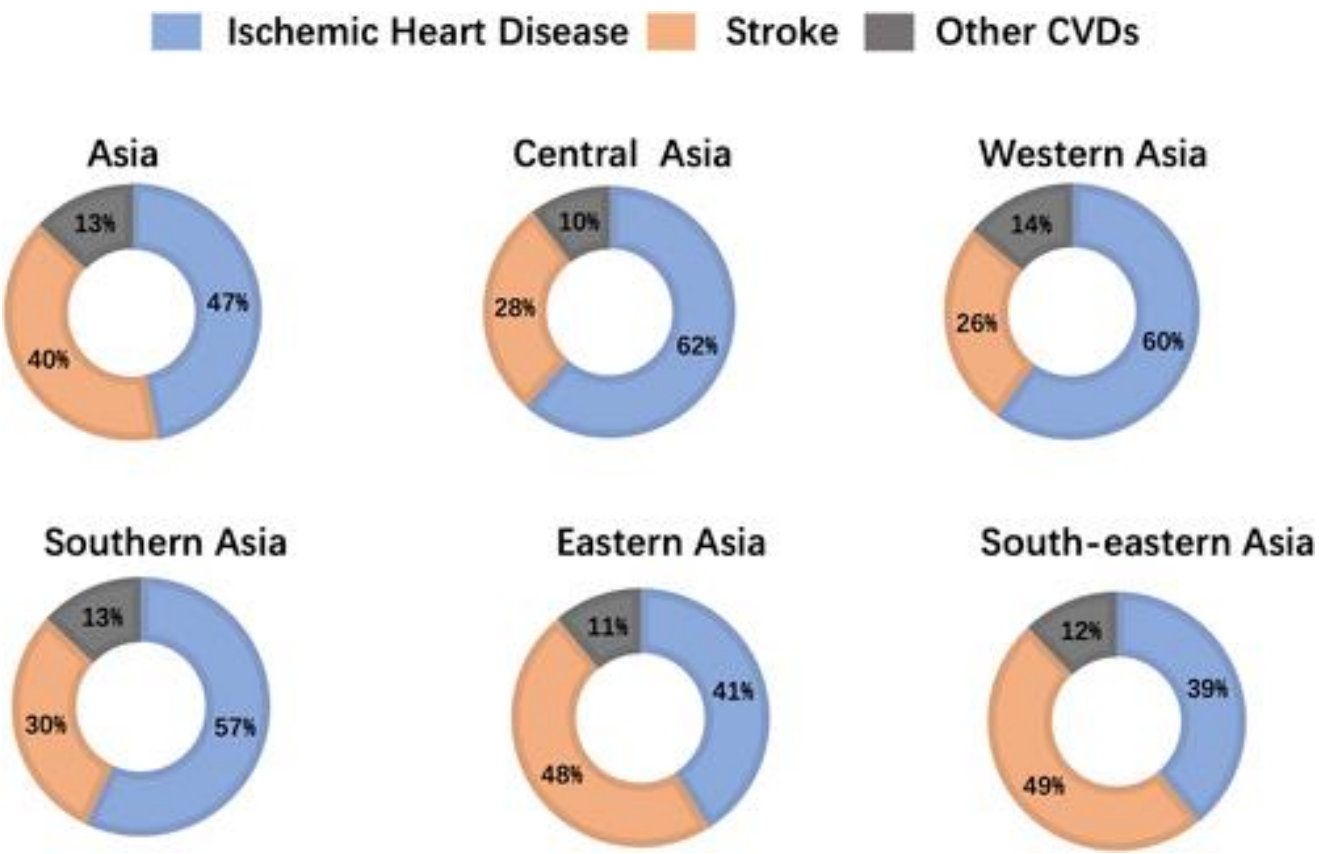


FIG : 2.4.1

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development of the wireless ECG monitoring system followed a systematic engineering approach, divided into key phases:

1. Requirement Analysis: Defined functional requirements such as signal acquisition, filtering, wireless transmission, and data display.
2. Component Selection: Chose cost-effective and reliable components including ECG electrodes, signal conditioning circuits, microcontrollers, and wireless modules.
3. Circuit Design and Prototyping: Developed an analog front-end circuit for ECG signal acquisition using filters and amplifiers. The design was prototyped on a breadboard and later moved to a PCB for stability.
4. Firmware Development: Wrote embedded code to sample ECG signals and transmit them wirelessly using Bluetooth/Wi-Fi.
5. Mobile Application Interface: Created a basic interface for real-time ECG waveform display using open-source tools.
6. Testing and Validation: Tested the system for signal accuracy, range, noise, and power consumption.

3.2 Tools and Technologies Used

Hardware Components:

- ECG Electrodes
- Instrumentation Amplifier (e.g., AD620)
- Filters (Low-pass and High-pass)
- Microcontroller (e.g., Arduino Uno or ESP32)
- Bluetooth/Wi-Fi Module (e.g., HC-05 or ESP32 inbuilt)
- Power supply (battery)

3.3 Solution Overview

The proposed system uses surface ECG electrodes to capture cardiac signals from the patient. These weak signals are amplified using an instrumentation amplifier and then filtered using bandpass filters to remove noise and artifacts. The conditioned analog signal is fed into a microcontroller with an ADC (Analog-to-Digital Converter), which samples the data and transmits it via Bluetooth or Wi-Fi to a smartphone or web-based monitoring platform.

3.4 Engineering Standards Applied

The following engineering standards were considered and partially applied:

- ISO 13485 – Quality management systems for medical devices (referenced for safe design practice).
- IEC 60601-1 – Basic safety and essential performance of medical electrical equipment (for electrical safety guidelines).
- IEEE 11073 – Personal Health Data Communication standards (for wireless data transmission protocols).
- ISO/IEEE 11073-10406 – Standard for basic ECG (applied in data formatting and transmission logic).

3.5 Solution Justification

Applying relevant engineering standards enhances the project's safety, interoperability, and reliability:

- Safety and Reliability: IEC 60601-1 guidelines helped in ensuring electrical isolation and noise minimization in the signal chain, critical for bio-signal acquisition.
- Data Communication: Adopting IEEE 11073 protocols ensures compatibility with standard health informatics systems and apps, improving usability and scalability**.**
- Quality Assurance: ISO 13485 principles guided the documentation and testing process, increasing confidence in device performance.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The developed wireless ECG system was tested for its ability to acquire, process transmit real-time ECG signals effectively. The following outcomes were evaluated:

- Signal Quality: The system successfully captured clear and distinguishable P, QRS, and T waves with minimal noise interference, confirming effective signal conditioning.
- Wireless Performance: Bluetooth/Wi-Fi modules maintained a stable connection within a 5–10 meter range, allowing real-time ECG monitoring on mobile devices without data loss.
- Cost Efficiency: The total component cost remained under ₹2000 (~$25), making it highly affordable compared to commercial ECG machines.

4.2 Challenges Encountered

- Component Compatibility: Interfacing the ECG circuit with microcontrollers required voltage level adjustments and careful PCB layout to avoid cross-talk.
- Mobile Interface Development: Ensuring real-time display and minimal latency on smartphones involved optimizing the app design and data transmission rate.

4.3 Possible Improvements

- Multi-lead ECG Acquisition: Expanding the system to support 3-lead or 12-lead ECG could provide more detailed cardiac analysis.
- Advanced Signal Processing: Integrating noise-reduction algorithms or AI-based arrhythmia detection can boost diagnostic accuracy.

4.4 Recommendations

- Clinical Validation: Collaborate with healthcare professionals to clinically test and validate the system against standard ECG machines.
- Waterproof and wearable design: Encasing the system in a wearable, weatherproof enclosure will improve durability and comfortable.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

1. Key Learning Outcomes:

1.1 Academic Knowledge

The project reinforced core concepts in electronics, biomedical instrumentation, and embedded systems. I applied theoretical knowledge—like bio-signal acquisition and wireless protocols—through hands-on implementation, bridging the gap between classroom and real-world applications.

1.2 Technical Skills

I gained practical experience in analog signal conditioning, microcontroller programming (Arduino, ESP32), circuit simulation (Proteus), PCB design (EasyEDA), and app development (MIT App Inventor), enhancing my proficiency with industry-relevant tools.

1.3 Problem-Solving and Critical Thinking

Facing challenges like ECG signal noise and wireless failures improved my troubleshooting skills and taught me to think critically and iteratively.

2. Challenges Encountered and Overcome

2.1 Personal and Professional Growth

Balancing academic and technical demands was tough, but it helped me build resilience and systematic documentation habits essential for research and industry.

2.2 Collaboration and Communication

Though primarily a solo effort, interactions with mentors and peers improved my ability to communicate technical issues, coordinate tasks, and accept constructive feedback.

3. Application of Engineering Standards

Standards like IEC 60601-1 and ISO 13485 guided safe design, noise reduction, and quality assurance. This experience introduced me to medical device compliance practices.

4. Insights into the Industry

The project revealed the complexity of developing low-cost, reliable, and user-friendly healthcare devices. It inspired my interest in biomedical and IoT-based medical systems.

5. Conclusion of Personal Development

This capstone project was a transformative learning experience. It deepened my technical knowledge, sharpened my problem-solving, and enhanced personal and collaborative skills, laying a strong foundation for a future in biomedical engineering.

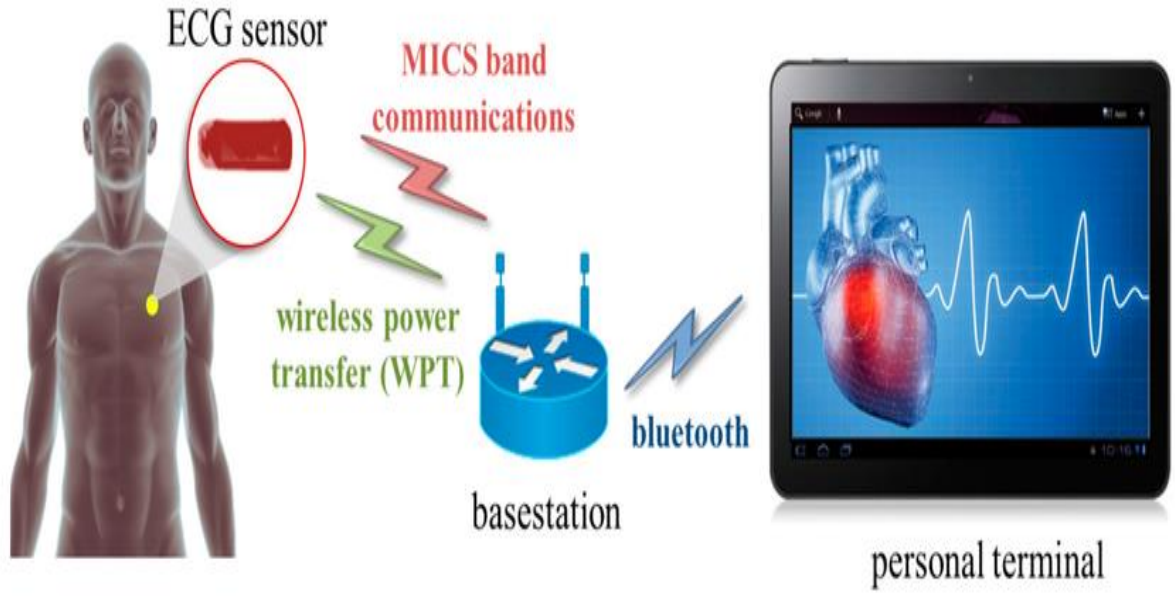


FIG : 5.1

CHAPTER 6

CONCLUSION

This capstone project successfully addressed the problem of limited access to affordable and continuous cardiac monitoring by designing a low-cost wireless ECG system using biomedical transducers. The solution involved acquiring ECG signals via electrodes, amplifying and filtering them, processing the data using a microcontroller, and wirelessly transmitting the signals for real-time display.

The project demonstrated that it is feasible to build a portable, low-cost ECG system suitable for use in home care, rural clinics, and resource-limited environments. The design aligned with relevant engineering standards to ensure safety, functionality, and interoperability. By doing so, the project not only met technical objectives but also contributed meaningfully to the growing field of remote and wearable healthcare technologies.

In conclusion, the project proved to be both technically viable and socially impactful. It offers a strong foundation for future research, development, and deployment of accessible health-monitoring devices and highlights the importance of applying engineering principles to real-world healthcare challenges.

REFERENCES

- World Health Organization. (2021). Cardiovascular diseases (CVDs).
- The Lancet. (2020). The rising burden of cardiovascular disease in South Asia. The Lancet Global Health, 8(5), e631. https://doi.org/10.1016/S2214-109X(20)30065-2
- Institute of Electrical and Electronics Engineers. (2021). IEEE 11073-10406-2011: Health informatics—Personal health device communication—Device specialization—Basic ECG (1- to 3-lead ECG).
- Indian Heart Journal. (2019). The role of portable ECG devices in early diagnosis and prevention. Indian Heart Journal, 71(2), 122–127. https://doi.org/10.1016/j.ihj.2019.04.005
- MIT App Inventor. (n.d.). Create apps for Android devices
- Patel, M., Shah, D., & Singh, R. (2020). Low-power wearable ECG system for real-time monitoring using BLE. Biomedical Signal Processing and Control, 60, 101954. https://doi.org/10.1016/j.bspc.2020.101954
- Chen, Y., Wang, X., & Zhang, H. (2019). Design of a compact ECG acquisition module for wireless monitoring. Sensors, 19(14), 3064. https://doi.org/10.3390/s19143064
- Zhang, Y., & Liu, X. (2018). Energy-efficient ECG monitoring system using IoT. IEEE Access, 6, 49152–49163. https://doi.org/10.1109/ACCESS.2018.2869018
- Gao, W., Emaminejad, S., Nyein, H. Y. Y., Challa, S., Chen, K., & Javey, A. (2021). Flexible biomedical sensors for real-time ECG detection. Nature Electronics, 4(9), 582–591. https://doi.org/10.1038/s41928-021-00625-7
- Shah, P., & Mehta, K. (2017). Evaluation of low-cost ECG sensor interfaces using Arduino. Procedia Computer Science, 115, 658–665. https://doi.org/10.1016/j.procs.2017.09.155

APPENDICES

Appendix A: List of Components

S. No	Component	Specification / Model	Quantity
1	Electrodes	Ag/AgCl Surface Electrodes	3
2	Instrumentation Amplifier	AD620 or INA128	1
3	Microcontroller	Arduino Uno / Pico / STM32	1
4	Low-Pass Filter Components	R = 10 kΩ, C = 0.1 µF	As needed
5	High-Pass Filter Components	R = 330 kΩ, C = 1 µF	As needed
6	Bluetooth Module	HC-05	1
7	Display	OLED Display / Serial Monitor	1
8	Power Supply	5V Battery / USB	1

Appendix B: Circuit Description

Signal Flow:

Electrodes → AD620 Amplifier → High-Pass Filter → Low-Pass Filter → Arduino ADC → Bluetooth Module → PC/Mobile Display

- Electrodes: Collect the ECG signal.
- AD620: Amplifies the low-voltage ECG signal.
- Filters: Clean the signal by removing low-frequency and high-frequency noise.
- Arduino: Converts analog ECG signal to digital.
- HC-05: Sends data to mobile or PC.
- Display: Shows ECG waveform (e.g., on Serial Plotter or mobile app

Appendix C: System flowchart

- Start

  ↓

- Electrodes capture ECG signal

  ↓

- Signal amplified using AD620

  ↓

- Noise removed using filters

  ↓

- Signal digitized by microcontroller

  ↓

- Data sent via Bluetooth module

  ↓

- ECG waveform displayed on PC/mobile

  ↓

- End